



RUOTOLO VINCENZO

Software/Robotic Engineer

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in vincenzo-ruoto Upwork

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Guadalajara, Mexico

ABOUT ME

Software Engineering looking for a full-time job in software developing, with experience in robotics backend/frontend and automotive (ADAS) software development. Proficient in C++ and Python, with a focus on high-performance solutions. Solid background in math, control systems, robotics, computer science, embedded systems, machine learning and IA.

WORK EXPERIENCE

Corporate

Luxoft (DXC) | Ford

Software developer and tester

03/2023 – ongoing Guadalajara (JA), Mexico

Engineered Python-based automation testing framework for Ford's BlueCruise (ADAS L2+), boosting validation coverage and reducing testing time.

Tasks:

- Design automated testing framework, including testing
- Review pull requests to ensure code quality

Luxoft (DXC) | BMW

Software developer

03/2022 – 03/2023 Turin (TO), Italy

Developed autonomous driving applications leveraging Adaptive AUTOSAR ARA::COM for inter-module communication.

Tasks:

- Implement production-grade features in modern C++, adhering to AUTOSAR architecture and MISRA coding guidelines
- Design and executed unit/integration tests

Freelance/Open Source

Open Source Contributor | python-can

Software Developer

2024 – 2025 Remote

- Implemented authentication middleware and contributed bug fixes to the python-can open-source repository
- Wrote unit tests improving test coverage
- Collaborated with maintainers via GitHub PRs

Cosmico

Software developer (Part-time)

01/2024 – 01/2025 Remote

Developed a cloud-native REST API backend with serverless components using FastAPI and SQLAlchemy, fully integrated with AWS (Lambda, API Gateway, ECR). Delivered a scalable, secure, and automated system for data processing and analytics with AI.

EDUCATION

Master's Degree in Mechatronic Eng.

Thesis: Optimal Trajectory Generation via MPC for a Four-Wheel Steering UGV

Politecnico di Torino

Bachelor's Degree in Aerospace Eng.

Thesis: Thales Alenia Space internship

Politecnico di Torino

COMPUTER SCIENCE

OS

Windows Ubuntu Kali

Programming languages

C++ C Python Matlab/Simulink
Bash LaTeX SQL NoSQL

Developer stack

ROS2 CMake Bazel Embedded Systems
REST API FastAPI SQLAlchemy Kafka
ML/AI Scikit-learn Pytorch

Tester stack

TDD pytest/unittest googletest

DevOps stack

Git GitHub/GitLab Ansible Docker
Agile/Scrum AWS

LANGUAGES

Italian (Native)

English

Spanish



Projects

Autonomous Robotic Systems

Project lead, Software developer

📅 01/2024 – Ongoing

📍 Remote

Leading the design, development, and deployment of a modular robotic architecture for UGVs, UAVs, and UUVs.

Tasks:

- Design system and module architecture with a focus on scalability and hardware abstraction.
 - Develop the Controller module for real-time robotic control and autonomy.
 - Deploy the architecture on embedded hardware platforms and managed the full application lifecycle.
 - Coordinate project roadmap, task allocation, and integration efforts.
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DRAFT

Software, DevOps and Virtualization squad leader

📅 10/2019 – 10/2023

📍 Turin (TO), Italy

Projects:

- Leonardo Drone Contest (10/2019 - 09/2021)
- European Space Agency - Space Resources Challenge (09/2021 - 09/2022)

Tasks:

- Design high-level robotics architectures.
- Develop navigation and control applications.
- Created and validated robotics simulations in Gazebo.
- Deliver embedded software.
- Created and validated robotics simulations in Gazebo.
- Led DevOps initiatives.